Journal of
ENGINEERING AND
TECHNOLOGY
MANAGEMENT
JET-M

J. Eng. Technol. Manage. 17 (2000) 1–37

www.elsevier.com/locate/jengtecman

Integration mechanisms and R&D project performance ☆

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Abstract

Information processing theory suggests the need for different types of integration mechanisms in R&D project management depending on levels of uncertainty and equivocality. This paper examines the use of these mechanisms and their links to project performance in a sample of 121 R&D projects in a large research laboratory. Overall, it is found that formal leadership, planning and process specification, and to a lesser extent information technology use are related to project performance while the positive effects of horizontal structures are apparently balanced out by their costs. The integration mechanisms studied act on performance partly through their effect on horizontal communications. Modest support was found for the contingency hypotheses derived from information processing theory. It appears that managers adjusted their use of horizontal structures, planning and process specification, and informal leadership to project uncertainty but not to project equivocality. The positive effects of horizontal communications on performance were found to be greatest under high project equivocality as would be predicted by information processing arguments. Moreover, with the exception of formal leadership, the use of integration mechanisms did not enhance performance in contexts of low uncertainty and low equivocality. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Information processing theory; R&D project management; Integration mechanisms; Innovation management

☆ The authors would like to thank the researchers and managers at Centerlab for their assistance in conducting this research. They are also grateful to Louis Lefebvre, Elizabeth Lefebvre, Prosper Bernard and Peter Wade for their support and encouragement. Finally, we thank the editor and four anonymous reviewers for their useful and pertinent comments on an earlier version of this article.

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PII: S0923-4748(99)00018-1